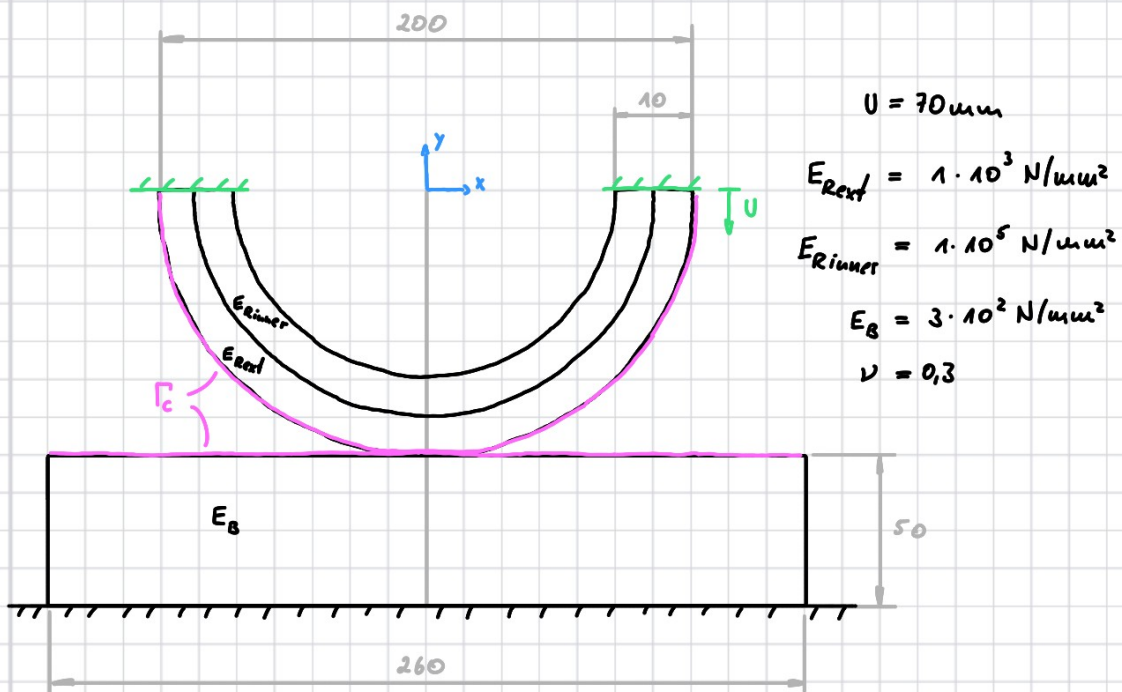
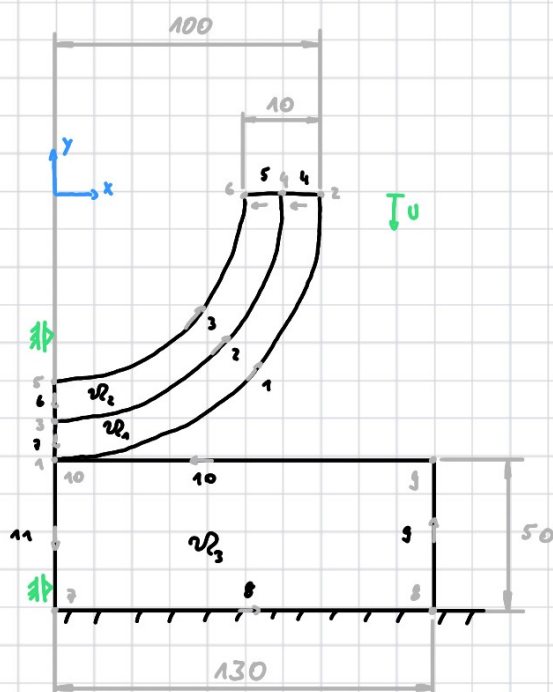


Contact finite strain



$U \rightarrow 140 \text{ steps}$
with $\Delta U = 0,5 \text{ mm}$



FE in CSx

$$\Psi = c_{10} (\bar{J}_1 - 3) + \frac{1}{D_1} (J - 1)^2 \quad \text{Neo-Hook} \quad \bar{J}_1 = J^{-2/3} \cdot J_1$$

$$\underline{\underline{F}} = \underline{\underline{J}} + \nabla u$$

$$\underline{\underline{F}} = \begin{bmatrix} F_{11} & F_{12} & 0 \\ F_{21} & F_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{plain strain}$$

$$J = \det \underline{\underline{F}} \quad c_{10} = \frac{E}{4(1+\nu)}$$

$$\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}} \quad D_1 = \frac{6(1-2\nu)}{E}$$

$$J_1 = \text{tr} \underline{\underline{C}}$$

$$\Psi_{tot} = \Psi_1 + \Psi_2 + \Psi_3$$

$$\Pi = \int_{\mathcal{R}} \Psi_{tot} dx$$

$$\mathcal{D}_\varepsilon \Pi[x] + \int_{\Gamma} \varepsilon_N g_N \delta g_N da + \int_{\Gamma} \frac{\varepsilon_N}{2} g_N^2 \delta(da) = R \stackrel{!}{=} 0$$

Contact algorithm

4 Gauss nodes ξ



$$\underline{x} = \underline{x}^* + \underline{u}$$

$$N_1 = 1 - \xi$$

$$N_2 = \xi$$

$$\underline{x}^s = N_1^s(\xi^s) \cdot \underline{x}_1^s + N_2^s(\xi^s) \cdot \underline{x}_2^s$$

$$\underline{a}^s = \underline{x}_2^s - \underline{x}_1^s$$

$$\underline{n}^s = \frac{1}{\|\underline{a}^s\|} \begin{bmatrix} a_y^s \\ -a_x^s \end{bmatrix}$$

$$\underline{x}^m = N_1^m(\xi^m) \cdot \underline{x}_1^m + N_2^m(\xi^m) \cdot \underline{x}_2^m$$

$$\underline{a}^m = \underline{x}_2^m - \underline{x}_1^m$$

$$\underline{n}^m = \frac{1}{\|\underline{a}^m\|} \begin{bmatrix} a_y^m \\ -a_x^m \end{bmatrix}$$

$$\xi^2 = (\underline{x}^1 - \underline{x}_1^2) \cdot \underline{a}^2 / (\underline{a}^2 \cdot \underline{a}^2)$$

$$(\underline{x}^1 - \underline{x}^2)^T \underline{n}^2 = g_N$$

$$\underbrace{(\underline{n}^2)^T}_{\underline{B}^T} \underline{H}_c \underline{x}_c = g_N$$

$$\underline{x}^1 - \underline{x}^2 = \underline{H}_c \underline{x}_c$$

$$= \begin{bmatrix} N_1^1 & 0 & N_1^2 & 0 \\ 0 & N_1^1 & 0 & N_1^2 \end{bmatrix} \begin{bmatrix} \underline{x}_1^1 \\ \underline{x}_2^1 \end{bmatrix} - \begin{bmatrix} N_1^2 & 0 & N_2^2 & 0 \\ 0 & N_1^2 & 0 & N_2^2 \end{bmatrix} \begin{bmatrix} \underline{x}_1^2 \\ \underline{x}_2^2 \end{bmatrix}$$

$$= \begin{bmatrix} \overbrace{N_1^1}^{\xi_1^1} & 0 & \overbrace{N_1^2}^{\xi_1^2} & 0 & -\overbrace{N_1^2}^{\xi_1^2} & 0 & -\overbrace{N_1^1}^{\xi_1^1} & 0 \\ 0 & \overbrace{N_1^1}^{\xi_1^1} & 0 & \overbrace{N_1^2}^{\xi_1^2} & 0 & -\overbrace{N_1^2}^{\xi_1^2} & 0 & -\overbrace{N_1^1}^{\xi_1^1} \end{bmatrix} \begin{bmatrix} \underline{x}_1^1 \\ \underline{x}_2^1 \\ \underline{x}_1^2 \\ \underline{x}_2^2 \end{bmatrix}$$

$$\underline{l}_1 = \|\underline{a}_1\|$$

$$\underline{t}_1 = \frac{\underline{a}_1}{\underline{l}_1}$$

$$\underline{A} = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\underline{L} = \underline{A}^T \underline{t}_1$$

$$\underline{F}_c = w \epsilon_N \left(g_N \underline{l}_1 \underline{B} + \frac{1}{2} g_N^2 \underline{L} \right)$$

$$\underline{K}_c = w \epsilon_n \left[l_s \underline{B}_c^T \underline{B}_c + g_n l_s \underline{H}_g + g_n (\underline{B}_c^T \underline{L}^T + \underline{L} \underline{B}_c) + \frac{1}{2} g_n^2 \underline{H}_c \right]$$

$$\underline{H}_c = \underline{A}^T \frac{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \underline{t}_s \otimes \underline{t}_s}{l_s} \underline{A}$$

$$\underline{H}_g = -\frac{1}{l_2} (\underline{v} \otimes \underline{w} + \underline{w} \otimes \underline{v}) - \frac{g_n}{l_2^2} (\underline{w} \otimes \underline{w})$$

$$l_2 = \|\underline{a}_2\|$$

$$\underline{t}_2 = \frac{\underline{a}_2}{l_2}$$

$$\underline{A}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

$$\underline{v} = \underline{H}_c^T \underline{t}_2$$

$$\underline{w} = \underline{A}_2^T \underline{n}_2$$

ACTIVE GAUSS NODES

for segment in slaves:

$\underline{x}_1, \underline{a}_1, \underline{n}_1$ from ξ_i^1

search master segment $\underline{x}_2, \underline{a}_2, \underline{n}_2$ and ξ_i^2

if $g_n \leq 0$: ξ_i^1 is active

search active segments

→ while True

Assemble $\underline{R} = \underline{R}_{int} - \underline{R}_{ext}$ and $\underline{K}$

for active ξ_i^1

get $\underline{x}^1, \underline{n}^1, \underline{q}^1$ and $\underline{x}^2, \underline{n}^2, \underline{q}^2$

$$\underline{K} += \underline{K}_c$$

$$\underline{R} += \underline{F}_c$$

apply bcs

$$\underline{K} \Delta \underline{u} = -\underline{R}$$

$$\underline{u} = \Delta \underline{u} + \underline{u}$$

if $\|\Delta \underline{u}\| < tol$: break

search active segments

if contact state changed
else next load step



$\hookrightarrow$

$$\text{active set} = \begin{bmatrix} n & i \\ n+1 & i \\ \vdots & \vdots \end{bmatrix}$$